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DEFENCE AND PEACE ECONOMICS: A TEN-YEAR RETROSPECTIVE

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INTRODUCTION

Defence economics is a relatively new subject area within the discipline of economics. In the year 2000, it will be 40 years since the publication of one of the pioneering texts in the field, namely, *The Economics of Defense in the Nuclear Age* (Hitch and McKean, 1960) and 10 years since the first issue of *Defence Economics*. The publication of the first journal in January 1990 was recognition that the field of defence economics was a recognized and acceptable specialism within the academic discipline of economics. Even though the Cold War ended over eight years ago with the demise of the Soviet Union on 20 December 1991, security concerns are still pervasive.[1] Ethnic-based civil wars have erupted in Europe, Africa, and the former Soviet republics over the last decade. Weapons of mass destruction (e.g. biological, chemical, and nuclear) can be acquired and used by rogue nations and terrorist groups, thereby creating security risks worldwide. Military downsizing in recent years brings its own challenges in terms of the need to strengthen ally ties and develop new generations of affordable weapons. The defence economics problem remains: limited resources

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[1]On these future challenges, see McGuire (1995), Sandler (1997), and Sandler and Hartley (1999).

require difficult choices under uncertainty. The necessity to apply state-of-the-art economic analysis to security and defence issues is every bit as important today as it was yesterday; thus, the journal continues to have crucial topics to analyse.

Subscribers to the journal are distributed worldwide and include universities, think tanks, research institutions, government agencies, defence contractors, practitioners, and members of the International Defence Economics Association (IDEA). In 1994 with volume 5, the journal's name was changed to *Defence and Peace Economics* to emphasize its interests in defence and conflict resolution issues.[2] There has been two changes in editors: Todd Sandler retired as the North America editor at the end of 1994 and was replaced by William Rogerson. In 1999, William Rogerson retired as editor and was replaced by Charles Anderton. These editorship changes have given the journal new interests and directions to supplement those of the founding editor, Keith Hartley. This latest change in editorship indicates the journal's commitment to publishing peace economics topics and increasing the flow of manuscripts from North America.[3] With this year's volume, the journal expands from four to six issues a year to accommodate the flow of manuscripts and to reduce the backlog of papers in the publishing queue.

The editors continue to be supported by an international Editorial Board which offers valuable advice and assistance with the refereeing of papers. Over the decade, there have been changes in the size and composition of the Editorial Board and the editors wish to acknowledge the contribution of the Board to the success of the journal. Similar acknowledgment is required to our many colleagues who give generously of their time to assist with the refereeing process.

The basic aims and scope of the journal remains the same: to embrace all aspects of the economics of defence, disarmament, conversion, and peace. Topic areas include the study of alliances and burden-sharing; military expenditures in developed and developing nations; arms races; terrorism; country surveys; the impact of disarmament on employment and unemployment; the prospects for conversion and the role of public policy in assisting the transition; the costs and benefits of arms control regimes; the trade in arms; the use of economic sanctions; and the role

[2]This title change also allowed the journal to shed its cumbersome subtitle: "The Political Economy of Defence, Disarmament and Peace."

[3]For the first five to six years, most manuscripts came from the United States. During the last four years, the bulk of manuscripts came from Europe.

of the United Nations in promoting security. Further topics involving economic efficiency include budgeting, the military production function, procurement policies, defence industries, collaborative weapons programmes, military manpower, internal markets in the armed forces, and public choice issues. These topics are investigated with the application of theoretical and empirical analysis. Theoretical tools include general and partial equilibrium analysis, game theory, comparative statics, constrained optimisation, and optimal control analysis. Concepts from industrial organisation, public economics, public choice, the new institutional economics, labour economics, macroeconomics, regional science, and international economics can be fruitfully applied to defence economic questions. Appropriate empirical tools involve both parametric and non-parametric statistical inference. Time-series analysis has, for example, been used to examine defence expenditure issues, arms races, and terrorism. Advanced econometric procedures have been applied to a host of military manpower issues (Warner and Asch, 1995). Virtually every econometric technique has been employed in the study of defence and peace economics.

The primary purpose of this paper is to examine and evaluate the coverage of the journal to date. In the process, we identify where this coverage has been either adequate or inadequate. It is our hope that the identification of these gaps in coverage, not only in the journal but also in the literature, will generate new articles on these topics for future issues. We conclude with some new directions for the journal.

DEFENCE AND PEACE ECONOMICS COVERAGE TO DATE

To date, the journal has published theoretical, empirical, conceptual and policy pieces on a wide array of topics using a variety of methodologies. This coverage can be put in perspective with the help of Table 1, which lists 21 topics covered in the first 10 volumes. For each topic, the number of separate articles published in *Defence and Peace Economics* (DPE) is given in parentheses. To place articles in categories, we exercised judgment, particularly when an article dealt with multiple sub-areas. In these instances, articles were assigned based on their primary focus; in a few cases, it was difficult to make the assignment. The authors to the articles are displayed in each sub-area in alphabetical order. If an author or set of co-authors contributed more than one paper, then the number of these articles follows the name.

Table 1 Coverage in *Defence and Peace Economics*, Volumes 1–10 (1990–99), by Sub-area

Defence expenditure studies[a] (29)[b]

Adams and Ciprut
Avramides
Ayanian
Beck
Dunne and Ron Smith
Ferreira and Barros
Finlay and Parker
Fontanel and Hébert
Fontanel and Ron Smith
Gold
Hartley
Hildebrandt
James, Solberg and Wolfson
Jones-Lee (2)[c]
Kammler
Kinsella (2)
Mohammed
Olszewski, Payne and Ross
Payne and Ross
Refeness, Kollias and Zapranis
Scheetz
Schiff, Gupta and Clements
R. William Thomas, Stekler and Glass
Waller
Ward and Park

Procurement processes and policies (21)

Bös
Bowlin
Cavin
Hall and Markowski (3)
Hall, Markowski and Thomson
Hartley
Hartley and Martin
Heyes and Liston-Heyes
Keating
Kirkpatrick
Liston-Heyes (2)
Manohara
Markoswski
Martin, Hartley and Cox
Matthews
Molas-Gallart
Pugh
Rogerson (2)

Economics of alliances (18)

Anderton
Ayania
Boyer
Bruce
Feng
Hansen, Murdoch and Sandler
Hilton and Vu
Ihori
Jones
Khanna and Sandler (2)
Linster
McGuire
Oneal
Palmer and Souchet
Wolfson
Wolfson and Robert Smith
Wong

Conversion (17)

Aben and Daures
Anthony
Bougrov
Brunat
Brunat and Malakhov
Cohen and Ward
Fontanel, Borissova and Ward
Fontanel, Samson and Spalanzani
Gonchar and Wulf
Hébert and De Penanros
Hooper and Butler
Kapstein
Le Nounail and Sauvin
Lock and Voss
Shkoratov and Fontanel
Ward, Cohen and Betsill
Zverev

Table 1 *Continued.*

Defence industries and industrial base (16)

Brzoska
Chakrabarti, Glismann and Horn
Dirksen
Dunne
Dussauge and Garrette
Hedvall
Lerner
Lovering
Martin and Hartley
Martin, White and Hartley
Rather and Celia Thomas
Taylore
Trevino and Higgs
Udis and Maskus
Walker and Willett
Willett

Military manpower (14)

Anderson, Halcousis and Tollison
Berner and Daula
Buddin
Downes
Kerstens and Meyermans
Martin
Mehay
Melese, Blandin and Fanchon
Owen
Marvin Smith
Solnick, Henderson and Kroeschel
George Thomas and Kocher
Warner and Asch
Withers

Country surveys (11)

see Table 2

Terrorism, guerrilla warfare, and insurgencies (11)

Brito and Intriligator
Enders, Parise and Sandler
Enders, Sandler and Cauley
Grossman
Grossman and Han
Lichbach
Nelson and Scott
Sandler
Scott
Shahin and Islam
Wilkinson

Defence and growth (9)

Alexander (2)
Atesoglu and Mueller
Huang and Mintz (2)
Kollias
Madden and Haslehurst
Mueller and Atesoglu
Murdoch, Pi and Sandler

Public choice and defence (8)

Anderson and Tollison
Brauer
Cowen and Lee
Goff and Tollison
Higgs
Higgs and Kilduff
Jones
Lee

Regional economics and defence (7)

Barber
Braddon
Braddon and Dowdall
Hooper and Cox
Parai, Solomon and Wait
Solomon
Wing

Table 1 *Continued.*

Defence R&D, technological change, and weapon development (6)	
Buck, Hartley and Hooper	Lichtenberg
Cavin	Molas-Gallart
Egea	Weston
Military posture and strategic policy (6)	
Cooper	Kuenne
Franck and Hildebrandt	Pickard
Hooper and Uttley	Sandler
Arms races (6)	
Anderton	Linden
Brito and Intriligator	Seiglie
Kollias and Makrydakis	Wolfson
Arms Trade (5)	
Brzoska	Levine, Sen and Ron Smith
Grobar, Stern and Deardorff	Kinsella
Levine, Mouzakis and Ron Smith	
Disarmament and arms control (5)	
Fontanel	Intriligator
Fontanel and Ward	Richards
Hoon	
Others (5)	
Anderson	Hedvall
Conybeare	Kirkpatrick
Coulomb	
Military production function (4)	
Cekota	Lovell, Morey and Wood
DeBoer and Blackley	Ridge and Ron Smith
Nature of defence economics (4)	
Intriligator (2)	Tinbergen
Reppey	
Defence budgets (3)	
Aufrant	Weidenbaum
Fontanel	
Developing countries and defence (2)	
Brauer	Landau
(but also see defence and growth)	

Notes: (a)This category includes papers concerned with macroeconomic consequences of defence spending, time series properties of defence spending, and the demand for defence. (b)Indicates number of articles in a category. (c)Indicates the number of articles by same author. Table based on all articles published in the journal up to, and including, Vol. **10**(2), 1999.

Defence expenditure studies had the most coverage, and included articles on the demand for defense spending, the macroeconomic consequences of defence spending, and the time-series properties of defence spending and related series. The next greatest coverage involved studies of weapon procurement, the economics of alliances and conversion. Procurement studies addressed contracting, management systems, procurement practices, financial aspects, and cost considerations. Given the large cutback in defence spending worldwide after the Cold War, it is not surprising that the conversion of defence industries and its economic consequences have generated so much research. Conversion in Russia has been of particular interest. Alliance studies in *DPE* examined aspects of burden sharing, free riding, and allied defense demands. With changes in the NATO membership, this interest should continue.

Coverage of defence industries and the defence industrial base has also been adequate, and addressed profitability, policies (e.g. offsets), collaborative programmes, and competitiveness. *DPE* has also published a variety of manpower studies concerning quit rates, reserves, recruitment strategies, compensation practices, volunteer forces, and other issues.

Moderate coverage characterizes country surveys; terrorism, guerrilla warfare, and insurgencies; defence and growth; public choice issues; and regional economic considerations. In Table 2, a list of country surveys, their authors, and their publication cite and date are given. Thus far, five NATO allies – West Germany, Greece, Spain, Turkey, Poland – have been surveyed, with surveys in progress for four other NATO allies – Italy, France, the United Kingdom, and the United States. Three countries – India, Pakistan, Sri Lanka – on the sub-continent have been studied and there are forthcoming surveys on Japan and South Africa. Surveys on China, Russia, Taiwan, South Korea, and Iran would be of interest, as would surveys on the other two NATO entrants and prospective NATO entrants (e.g. Slovakia and Romania). *DPE* studies on terrorism and insurrections have employed a variety of empirical methods to evaluate policy effectiveness. General equilibrium analysis has been applied to study the effects of foreign aid and other decision variables on insurgencies. In a series of papers, the Feder-Ram, supply-side model has been used to investigate the impact of defence spending on growth (see Ram, 1995). Some of these articles have determined whether or not military cutbacks would lead to a peace dividend from improved growth. The cross-sectional, Feder-Ram studies possessed more robust empirical results than the single-country

Table 2 List of country surveys

Country surveyed	Author(s)	Volume (Issue)
India	Michael D. Ward, David Davis, Mohan Penubarti, Sheen Rajmaira and Mali Cochran	3(1), 1991
Sri Lanka	Lisa Morris Grobar	3(2), 1992
West Germany	Helmut Maneval	5(3), 1994
Pakistan	Ron Matthews	5(4), 1994
Poland	Maciej Perczynski, Pawel Wieczorek and Katarzyna Zukrowska	6(1), 1995
Australia	Stefan Markowski and Peter Hall	6(2), 1995
Greece	Christos Kollias	6(4), 1995
Yugoslavia	Miodrag Ivanovic	7(3), 1996
Spain	Jordi Molas-Gallart	8(3), 1997
Turkey	Selami Sezgin	8(4), 1997
Israel	Michael Beenstock	9(3), 1998

Note: Surveys in progress are on Japan, the United Kingdom, the United States, France, and Italy.

studies (Murdoch, Pi and Sandler, 1997). Public choice studies concerned such topics as the effects of public opinions on defence spending, bureaucratic-motivated defence spending, and Abraham Lincoln's troop deployment strategy to maximize his electoral college vote in the 1864 election (Anderson and Tollison, 1991). Regional economics articles investigated defence spending influences on regional employment, prosperity, and growth.

Areas with relatively little coverage includes the remaining topics in Table 1. We find surprising the small number of articles on arms races, arms trade, disarmament, and developing countries. There are clearly important concerns involving arms races even in the post-Cold War era – e.g. regional arms races (India and Pakistan, China and Taiwan, North and South Korea) and the proliferation of weapons of mass destruction. With reduced military budgets in the major arms-producing nations, weapons sales are a means of increasing production runs so as to lower unit costs by spreading fixed costs of R&D over more units. Unfortunately, these sales create negative externalities if conflicts were to develop that then threaten allies and others with these traded armaments. Paul Levine, Ron Smith, and colleagues have been pursuing an important line of research on these externalities and related arms sales issues (Levine, Sen and Smith, 1994).

Although *DPE* has published a large number of articles on the conversion of defence industries, it has published relatively little about the

stability, dynamics, and efficiency aspects of disarmament. From a strategic viewpoint, it is important to gain insight as to what can break arms race dynamics associated with Prisoner's Dilemma kinds of games (Sandler and Hartley, 1995, ch. 4). If this dynamic were better understood, then organisations such as the United Nations could make more informed decisions to address arms races in developing countries. In fact, there are a host of issues concerning developing countries that go beyond the growth and defence spending question, and have received scant attention in *DPE*. These concerns include technology transfers through arms trade, the development of arms industries, the implication of defence on trade imbalances, and the impact of defence spending on income distribution.

From time to time, *DPE* has commissioned guest editors to put together special issues on specific topics. These special issues fulfil two purposes: (i) address a topic of current importance, and (ii) rectify a perceived shortfall in coverage. Sometimes these special issues have resulted from a conference for which the best papers have been revised and later published in a single issue.[4] In Table 3, special issues published

Table 3 Special issues: Topics and Editors

Topic	Editor(s)	Volume (Issue)
Defence human resources	Glenn Withers	2(4), 1991
Terrorism, guerrilla warfare, and insurrections	Todd Sandler	3(4), 1992
UK and European defence industrial base	Susan Willett	4(2), 1993
Disarmament	Keith Hartley	5(2), 1994
Conversion	Jacques Fontanel, Ivan Samson	6(3), 1995
Regional dimension of defence economics	Derek Braddon	7(1), 1996
Military expenditure in industrial democracies	Hans Kammler	8(1), 1997
International defence procurement	Stefan Markowski, Peter Hall	9(1–2), 1998
Conversion of the Russian defence industry	Eric Brunat	9(4), 1998
Forthcoming		
Security in S. Asia	M. Chatterji	
Defence in Turkey and Greece	S. Sezgin	

[4] This was the case for the special issue on military expenditure in industrial economies and the issue on the conversion of the Russian defence industry.

in the first ten volumes of *DPE* are listed along with their editor(s), volume placement, and date. Special issues in progress include those on corporate governance in the arms industry, security and South Africa, security issues in South Asia, defence and the history of economic thought, and economic aspects of defence in Turkey and Greece.

GAPS IN THE JOURNAL COVERAGE

There are a number of areas of defence economics where the journal would be particularly interested in publishing high-quality papers. One area is game-theoretic analysis of defence. Even though game theory has become a ubiquitous tool of economic analysis and political science during the past decade, *DPE* has published relatively few articles using these techniques. Noteworthy exceptions are papers by Bruce Linster (1993) on rent seeking in alliances, William Rogerson (1991) on procurement games, and Neil Bruce (1990) on allied and adversarial relationships. Thus far, non-cooperative game theory, where agents act strategically and independently of one another, has been used almost exclusively in any game-theoretic analysis of arms races, alliances, procurement, and terrorism. Cooperative game theory, where players form coalitions to share benefits from coordinating actions, holds promise as a method to gain further insights on these and other topics of defence economics. For example, the theory of the core[5] is being used to explain alliance formation and expansion in recent work (Sandler, 1999). Game theory is also applicable to the study of framing, ratifying, and verification of arms-limitation treaties. Once verification is examined, issues of asymmetric information, in which one side is better informed than another, becomes relevant. The intense strategic interaction between terrorists and the authorities is also worthy of further analysis with the aim of producing a theory explaining terrorist campaigns.

Another gap in *DPE* coverage concerns peace economic studies on peacekeeping missions, regional arms races, and conflict resolution. NATO's new strategic doctrine increases the importance of crisis-management missions throughout Europe and in regions of economic and strategic value to NATO. Since the start of the 1990s, UN peacekeeping operations have increased in numbers and importance (Sandler and

[5] The core exists when there are no blocking coalitions that can form and improve their well-being.

Hartley, 1999), resulting in greater amounts of resources being allocated directly and indirectly (e.g. developing power projection capabilities). Further theoretical and empirical studies of peacekeeping and related issues are of interest. Regional arms races between and within countries (i.e. between rival factions) may increase the future need for peace enforcement operations.

Although *DPE* has published a number of articles by political scientists (e.g. Michael Ward, Patrick James, Paul Wilkinson, John Oneal, Glenn Palmer and Mark Boyer), there is an interest to increase these contributions to bolster the journal's interdisciplinary orientation and political economy focus. The publication of the interesting article by Patrick James, Eric Solberg and Murray Wolfson (1999) on whether or not democratisation promotes peace is intended to attract more political science contributors. *DPE* should be viewed as an available forum for controversies and contributions involving theoretical, empirical, and policy papers on defence and peace topics from political science and related disciplines. The journal's expansion to six issues annually will allow it to accommodate these papers.

There are also opportunities for policy-makers and practitioners to publish articles on policy problems. There have been a few examples from officials in Defence Ministries (e.g. Cooper, 1990; Weston, 1996) and some contributions from retired staff (Kirkpatrick, 1995; Pugh, 1993); but there have been no contributions from industry nor from the armed forces. Nor have there been many critical commentaries on published articles. These are welcomed as a means of advancing knowledge and understanding.

The editors would also welcome good quality case studies of defence industries, projects and of conversion (cf the pioneering contribution of Peck and Scherer, 1962). Since the end of the Cold War, there have been massive changes in the size and structure of defence industries reflected in job losses, plant closures and mergers with economic impacts on local economies. There is scope for good, in-depth analytical, empirical and critical evaluations of national defence industries, including the component sectors of aerospace, electronics, land systems (armoured fighting vehicles and ordnance), nuclear systems and shipbuilding. Problems arise in defining defence industries, in obtaining data and in identifying the extent of supply chains for the different defence industries (Hartley *et al.*, 1997). Disarmament following the end of the Cold War has created major challenges for nations wishing to retain a national defence industrial base. These industries now face fewer new projects, smaller production orders and gaps in both development and production

work which makes it increasingly difficult and costly to maintain some "key" defence industries (e.g. nuclear systems). The journal welcomes case studies of how different countries have addressed these industrial base issues (e.g. shut-down and mothballing options; their costs and time-scales for restarting production).

There is also a need for in-depth economic evaluation of major procurement programmes (e.g. US F–22; the four nation Eurofighter; aircraft carriers; nuclear-powered submarines). These are costly projects characterised by cost overruns and delays and they need to be evaluated critically with the aim of identifying lessons for future procurement policy (e.g. type of contracts; role of competition). Conversion is another area for the case study method. Despite the substantial number of articles on the subject, there remains scope for detailed economic studies of conversion identifying successful and failed conversions and the reasons for success or failure. Associated studies of labour markets for redundant military personnel and redundant defence workers (including scientists) would provide important insights into the operation of labour markets and the extent to which military and defence industry skills are transferable to the civil sector.

The military production function remains relatively unexplored. Few articles have addressed the challenges of measuring *defence output* and the associated valuations placed on the protection of human life and its quality (e.g. peace, security, democracy: Jones-Lee, 1990). For NATO nations following the end of the Cold War, the definition of defence output has changed to embrace peace-keeping, peace-making, crisis management and humanitarian relief both within and outside the traditional NATO boundaries. These new interpretations of defence output have resulted in a focus on proxy measures of final output in such forms as defence capabilities (e.g. the capability of fighting two regional conflicts or being involved in two to three Bosnia-type peace-keeping operations).

The *input side* of the military production function raises similar challenges for researchers. There is scope for modelling the armed forces as non-profit organisations, for assessing their efficiency in converting inputs into outputs and for examining substitution possibilities, including the impact of new technology, and their implications for the traditional monopoly property rights of each of the armed forces (e.g. surface-to-air missiles, cruise missiles and unmanned air vehicles operated by the army versus manned fighter and bomber aircraft operated by the air force). There are opportunities for introducing or simulating market-type solutions for the armed forces, for private contractors to

replace "in-house" functions traditionally undertaken by the armed forces (e.g. repair, maintenance and training) and for private finance solutions whereby the armed forces instead of owning equipment or facilities would 'lease' them from private industry. For example, instead of the armed forces providing their own aircraft, bases and personnel for flying training, they could allow private industry to undertake all flying training in return for the payment of a fee per qualified pilot. There is considerable scope for papers reporting and evaluating international experience on the military production function and alternative solutions to promoting efficiency.

FUTURE DIRECTIONS AND CONCLUDING REMARKS

DPE is launching a literature survey series that focuses on key issues in theory, policy, or application. Surveys could, for example, address: empirical arms race studies; the economics of appropriations; the political economy of revolutions; alternative methods of measuring security; the economics of peace and conversion; and the economics of peace-keeping and crisis management. The journal is especially interested in surveys on topics not covered in the *Handbook of Defence Economics*, or in new developments, or on significant updates to topics surveyed in the *Handbook* (Hartley and Sandler, 1995). Surveys addressing applications and policy issues are particularly welcome.

The journal is adding some additional topics to its aims and scope. With the publication of Fanny Coulomb's (1998) paper on Adam Smith, *DPE* is interested in papers on the history of thought and defence economics. Other new topics include the economics of modern warfare and the revolution in military affairs, manpower studies on rapid deployment forces, the political economy of overlapping alliances, the economics of weapons of mass destruction and the costs and burden-sharing arrangements for new military missions undertaken by NATO, the European Union and the UN.

The current anniversary issue celebrates the journal's past decade by turning to some of the most influential defence and peace economists. These authors have been asked to give a brief overview of topics of current importance to defence and peace economics. Each article pays special attention in identifying new directions and unanswered questions. Many of the contributions to the first issue of *DPE* are also contributors here; these include Paul Dunne, Michael Intriligator, Martin McGuire, Ron Smith, Keith Hartley and Todd Sandler.

McGuire explores the future agenda for defence economics in an era likely to be characterised by a "staggering upheaval" in the functions of military forces. He addresses such issues as alliances and the avoidance of big wars, the limits of disarmament, the management of lesser conflicts, defence and development and he forecasts that defence economics will become part of political economy. There follows two papers on arms races, which has been a major focus of theoretical and empirical work in defence economics. Smith, Dunne and Nikolaidou assess the technical difficulties involved in estimating arms race models and conclude that arms race models are much less mechanical than the traditional action–reaction model suggests. Intriligator and Brito argue that future arms races will be different from the Cold War arms race and they forecast "even greater instabilities." New theoretical developments will be based on increasing returns to scale in production, new technology and recent developments in game theory. Anderton presents a new dimension of defence and peace economics with implications for the core of the economics discipline. He argues that when conflict is introduced into the production-exchange economy, "the traditional economics landscape takes on dramatically new shapes."

The economics of military alliances has formed one of the foundations of defence economics. Okamura develops this literature by presenting a multiple public good model which explores the effect of comparative advantage in providing public goods in the form of defence and foreign aid. The model is applied to the USA–Japan alliance and it confirms that the United States has a comparative advantage in supplying defence and Japan in foreign aid. Warner and Asch show how past research in defence manpower economics continues to provide relevant guidance to policy-makers addressing the new technological and strategic environment. The future agenda for defence manpower economics will embrace the capital-labour requirements of new military missions for US forces (e.g. peace-keeping; anti-terrorism), the role of reserve forces, the opportunities for contractorisation by the private sector, the impact of demographic trends on the supply of military personnel (e.g. higher education) and optimal compensation policies. This anniversary issue concludes with Fontanel and Coulomb analysing economic aspects of disarmament and the future of disarmament. They start from "old-fashioned" disarmament analysis and outline a new economics of disarmament including the Russian economic crisis, the maintenance of economic wars and globalisation. All these contributions confirm that defence and peace economics has "grown-up" and is now an established part of the discipline of economics. The next century

offers an exciting and challenging prospect for new theoretical, empirical and policy work in the field of defence and peace economics.

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